

Group 1 :

MoSCoW

User Requirements Specification(URS) :

FR.1(Smart Zonal Storage Optimizer)[must] : "The system shall suggest the most suitable storage location for each inventory item based on how frequently it is sold and its physical size"

FR.3(IoT Weight Sensor Simulation)[must] : "The system shall automatically detect when the actual weight of items on a shelf does not match the expected weight based on recorded inventory, and shall alert the warehouse manager of any discrepancy."

FR.2(Volumetric Capacity Calculator)[should] : "The system shall prevent warehouse staff from assigning more stock to a zone or aisle than it can physically hold, by calculating the total volume of current stock against the available bin capacity

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FR.6(Inventory "Reserved" vs. "Available")[should] : "The system shall ensure that items added to a customer order are immediately locked and cannot be sold to another customer, displaying only truly available stock to users at all times

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FR.9(Inventory Cycle Count Scheduler)[Good] : "The system shall automatically identify and flag specific warehouse aisles that are due for a manual stock audit, based on how long it has been since their last check or whether they have a history of stock discrepancies"

System Requirements Specification :

FR.1(Smart Zonal Storage Optimizer)[Must] :

"The system shall calculate a velocity_score for each InventoryItem based on units sold in the last 30 days."

"The system shall classify each item as Zone A, Zone B, or Zone C based on its velocity_score against predefined thresholds."

"The system shall check available bin capacity within the classified zone before assigning a storage location."

"The system shall assign the item to the best-fit bin and update the storage_location field upon manager confirmation."

"The system shall display the suggested location to the Warehouse Manager on the inventory dashboard."

FR.3(IoT Weight Sensor Simulation)[Must] :

"The system shall calculate expected_weight for each shelf by multiplying unit_weight by quantity_on_hand for all assigned items."

"The system shall compare expected_weight against the simulated current_shelf_weight after every stock movement."

"The system shall flag a StockDiscrepancy record when the variance exceeds $\pm 2\%$ tolerance."

"The system shall display active discrepancy alerts to the Warehouse Manager with shelf ID, item, and variance amount."

"The system shall trigger an audit task for the affected aisle when a discrepancy is flagged."

FR.2(Volumetric Capacity Calculator)[Should] :

"The system shall calculate the occupied volume of each bin by summing length × width × height for all items currently assigned to it."

"The system shall reject any new stock assignment where the incoming item's volume would exceed the bin's max_capacity_cm3."

"The system shall display a capacity warning to the Warehouse Manager when any zone reaches 80% occupancy."

"The system shall suggest the next available bin with sufficient remaining capacity when a rejection occurs."

FR.6(Inventory "Reserved" vs. "Available")[Should] :

"The system shall maintain quantity_available and quantity_reserved fields for each InventoryItem."

"The system shall decrease quantity_available and increase quantity_reserved when an item is added to a customer order."

"The system shall display only quantity_available to Floor Staff and the Supplier portal."

"The system shall release quantity_reserved back to quantity_available if an order is cancelled or times out after 30 minutes."

"The system shall permanently deduct quantity_reserved from total stock upon confirmed order dispatch."

FR.9(Inventory Cycle Count Scheduler)[Good] :

"The system shall record the last_audit_date for each aisle upon completion of every cycle count."

"The system shall flag an aisle for audit when 30 days have elapsed since its last_audit_date."

"The system shall flag an aisle for audit when its discrepancy_count exceeds 3 within the last 30 days."

"The system shall generate a prioritized audit queue ordered by discrepancy count and last audit date."

"The system shall display the audit queue to the Warehouse Manager with aisle ID, days since last check, and discrepancy count."

"The system shall assign the audit task to an available Floor Staff member and update the aisle status to Audit Pending."